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Department of Computer Science

CS11021– Programming Laboratory (Intake 43)

Lab Sheet 4

1. Write a C++ program to **convert a temperature** from **Celsius to Fahrenheit** using the equation:

$$F = \left(\frac{9}{5} \times C \right) + 32$$

- Take **Celsius temperature** as input.
 - Use **casting** to ensure floating-point division.
 - Display result with **2 decimal places**.
2. Write a program that calculates **Body Mass Index (BMI)** using:

$$BMI = \frac{weight(kg)}{height(m)^2}$$

- Input **weight (kg)** and **height (m)**.
- Format output with **precision up to 2 decimal places**.

3. The force of gravity between two objects is given by:

$$Speed = \frac{Distance}{Time}$$

Input **distance (in meters)** and **time (in seconds)**.

Display **speed** using **scientific notation**.

4.

The force of gravity between two objects is given by:

$$F = G \times \frac{m_1 \times m_2}{r^2}$$

where $G = 6.674 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$.

- Input m_1 , m_2 (masses in kg) and r (distance in meters).
- Format output with **scientific notation**.

5.

The area of a triangle is given by:

$$A = \frac{1}{2} \times base \times height$$

- Input **base** and **height**.
- Ensure proper division using **casting**.
- Display **formatted output** with **width()**.

6.

Calculate the **distance** between two points (x_1, y_1) and (x_2, y_2) :

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- Input x_1, y_1, x_2, y_2 .
- Use **precision formatting**.

7.

The **kinetic energy** of an object is:

$$KE = \frac{1}{2}mv^2$$

- Input **mass (kg)** and **velocity (m/s)**.
- Use **casting for division**.

8. Write a program that takes **student scores** and calculates:

- **Total marks**
- **Average (2 decimal places)**

```
Student: Mike
Math: 85
Science: 90
English: 78
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Total:    253
Average:  84.33
```

9.

For males:

$$BFP = (1.20 \times BMI) + (0.23 \times Age) - 16.2$$

For females:

$$BFP = (1.20 \times BMI) + (0.23 \times Age) - 5.4$$

- Input weight, height, age, and gender.
- Display formatted output.

Part II

Question 10.

An electricity provider calculates a household's monthly electricity bill based on the total number of units consumed during the month. Each unit has a fixed cost decided by the provider. At the end of the billing period, the total consumption and cost per unit are recorded.

The total electricity bill is obtained by multiplying the number of units consumed by the cost per unit. The final bill amount must be displayed in a user-friendly format.

Type a C++ program to calculate the monthly electricity bill.

Question 11.

During a long-distance journey, a vehicle's distance traveled and the total fuel consumed are measured. Using these values, the vehicle's fuel efficiency is calculated to understand how efficiently the vehicle uses fuel.

Fuel efficiency is defined as the ratio between the distance traveled and the fuel used. Since both values may contain decimal points, appropriate data types and type casting should be used to ensure accurate calculation.

Type a C++ program to calculate and display the fuel efficiency of the vehicle with suitable formatting.

Question 12.

In a university course, student attendance is monitored throughout the semester. The total number of working days and the number of days a student attended classes are recorded at the end of the semester.

The attendance percentage is calculated by dividing the attended days by the total working days and converting the result into a percentage value. Care must be taken to avoid integer division during the calculation.

Type a C++ program to compute and display the student's attendance percentage in a properly formatted manner.

Question 13.

When downloading a digital file from the internet, the total file size and the average download speed determine how long the download will take. File size is usually measured in megabytes (MB), while download speed is measured in megabits per second (Mbps).

To calculate the download time in seconds, the file size must first be converted into bits and then divided by the download speed. Accurate type casting and formatted output are required to display the result clearly.

Type a C++ program to estimate and display the download time.

Question 14.

A mobile service provider charges users based on their monthly internet data usage. At the end of the month, the total amount of data used by a customer is recorded in megabytes (MB). The provider also specifies a fixed cost per gigabyte (GB) of data usage.

The total bill is calculated by first converting the data usage from megabytes to gigabytes and then multiplying it by the cost per gigabyte.

Type a C++ program to calculate the total monthly internet data usage cost.

Requirements:

- Take the total data used (in MB) as input.
- Take the cost per GB as input.
- Convert MB to GB using appropriate **type casting**.

- Calculate the total cost.
- Display the final cost **formatted to 2 decimal places**.